Optional Practice 2: How Heart Cells Work Together

Quantitative Physiology: *this practice is optional and requires no separate submission*.

How can many individual cells coordinate to help an organ pump blood? Build simple simulations of electrical communication and contraction. Then, if you want a greater challenge, connect activation to a pumping chamber.

Complete Parts 1 and 2 in any order. Part 3 is an optional extension. Use illustrative values for this educational simulation only.

Physiology background

The sinoatrial (SA) node contains specialized cardiac muscle cells that initiate the heartbeat. Nerves regulate heart rate; the SA node itself is not nervous tissue. Electrical connections called gap junctions allow neighboring muscle cells to communicate. Contractile muscle produces force, connective tissue supports the organ, and the inner lining helps form the blood-contacting surface.

These models focus on electrical communication, active contraction, and passive chamber mechanics. They do not represent every tissue or biological process in the heart.

Before coding

- Identify each variable, parameter, and unit. Explain what makes each variable increase or decrease.
- Predict the effect of changing one parameter before running the simulation.
- Use Gemini or another coding tool to help implement the model, then check the equations and results yourself.

Part 1 Electrical communication between cells

Represent a small piece of tissue as 10 cells connected in a line. Let V_i be the voltage of cell i. Only directly adjacent cells are connected.

$$C\,dV_i/dt = -gL(V_i - EL) + ggap\,\Sigma\,(V_neighbor - V_i) + I_i(t)$$

The sum is over connected neighbors. End cells have one neighbor. C is membrane capacitance; gL is leak conductance; EL is the leak reversal potential; ggap measures electrical coupling; $I_i(t)$ is applied current.

Illustrative setup

Use time in ms, voltage in mV, capacitance in pF, conductance in nS, and current in pA. Set C = 100 pF, gL = 1 nS, EL = −70 mV, and all initial voltages to −70 mV.

Stimulate only cell 1 with 20 pA for 50 ms, starting at 100 ms and repeating every 1,000 ms. Run for 3,000 ms. Compare coupling of 2 nS, 0.2 nS, and 2 nS with the connection between cells 5 and 6 removed.

Explore and explain

- Plot voltage against time for cells 1, 5, and 10. Add a heat map showing voltage across all cells over time.
- How do coupling strength and distance from the stimulus affect the size and timing of the voltage response? Explain any trade-off you observe.
- When the connection is removed, which cells still respond? Why?

Check your model

With no applied current and every cell at EL, all voltages should remain constant. With ggap = 0, unstimulated cells should remain at EL. Confirm that reducing the numerical time step or tightening solver tolerances does not substantially change the curves.

This is a passive electrical model: signals spread and weaken. It does not generate cardiac action potentials. The repeating input is a supplied stimulus, not a biological pacemaker model.

Part 2 Same cells different contraction times

Model 20 cells, each with an activation level a_i between 0 and 1. For this part, prescribe the activation commands directly; you do not need to convert Part 1 voltages into activation.

$$da_i/dt = (u_i(t) - a_i)/\tau$$

$$F_i(t) = F\max\, a_i(t) \qquad F\text{total}(t) = \Sigma_i\, F_i(t)$$

$u_i(t)$ equals 1 during a cell's activation command and 0 otherwise. τ sets the response time. Fmax is the maximum force of one model cell.

Illustrative setup

Use seconds. Set τ = 0.10 s, Fmax = 1 arbitrary force unit, and $a_i(0)$ = 0. Activate each cell once for 0.20 s. Run for 3 s so the contractions can relax.

- Coordinated case: all activation commands begin at 0.20 s.
- Scattered timing: distribute the 20 start times evenly from 0.20 to 0.60 s.
- Weaker cells: repeat the coordinated case, but reduce Fmax to 0.5 for five cells.

Explore and explain

- Plot several individual force curves and the total force for each case.
- Compare peak total force, time of the peak, and the area under the total-force curve.
- Does scattered timing necessarily reduce the area under the curve, or mainly change its shape and peak? Explain using the unchanged individual pulses.
- Why can this force sum not tell you directly how much blood the heart pumps?

Check and interpret

Verify that activation stays between 0 and 1 and returns toward zero after each command. For a constant command u, check $a(t) = u + [a(0) - u] \exp(-t/\tau)$. Exact simultaneous activation is a comparison case; the real heart activates in an organized sequence.

Part 3 Optional pumping chamber challenge

Extend the exercise to one simplified chamber. Blood enters through an inlet valve and leaves through an outlet valve. Keep upstream and downstream pressures fixed.

$$dV/dt = Qin - Qout$$

$$P(t) = E(t)[V(t) - V_0]$$

$$E(t) = Emin + (Emax - Emin)a(t)$$

$$Qin = \max[(Pin - P)/Rin, 0]$$

$$Qout = \max[(P - Pout)/Rout, 0]$$

V is chamber volume; P is chamber pressure; V_o is a reference volume. E describes pressure generated per unit volume above V_o. Emin represents resting stiffness; Emax represents activated stiffness. The resistances Rin and Rout oppose flow. The max function implements ideal one-way valves.

Illustrative setup

Use seconds, mL, and mmHg. Set V_o = 10 mL, V(0) = 100 mL, Pin = 10 mmHg, Pout = 80 mmHg, Rin = 0.05 mmHg·s/mL, Rout = 0.10 mmHg·s/mL, Emin = 0.08 mmHg/mL, and Emax = 2.0 mmHg/mL.

Generate chamber activation using $da/dt = (u - a)/\tau$ with τ = 0.05 s and a(0) = 0. Let u = 1 for the first 0.30 s of each 1 s cycle and 0 otherwise. Simulate 20 cycles and compare the last complete cycle.

Change one factor at a time

- Weaker contraction: reduce Emax from 2.0 to 1.0 mmHg/mL.
- Stiffer chamber at rest: increase Emin from 0.08 to 0.16 mmHg/mL, keeping Emax fixed.
- Faster activation: shorten the cycle from 1.0 to 0.60 s, keeping the activation command duration at 0.30 s.

Explore and check

Plot chamber volume, pressure, inlet flow, and outlet flow. Calculate ejected volume per cycle as the integral of Qout over one cycle. Multiply by cycles per minute to obtain the model's output per minute.

- Which changes affect filling, ejection, or both? Does a faster cycle always produce more output per minute in your model?
- Verify that volume change equals the integral of inflow minus outflow. Both valve flows must be nonnegative.
- Check whether successive cycles have settled before comparing them. Repeat with a smaller time step or tighter solver tolerances.

This chamber model omits spatial cell coordination, calcium dynamics, detailed valves, and the full circulation. E represents overall chamber mechanics, not connective tissue alone. Do not equate the force sum in Part 2 with pressure without an additional mechanical model.

Coding support and reflection

Optional starter prompt

Use the following prompt if you would like help getting started. Paste the equations, parameter values, and scenarios for the part you selected after the prompt.

“Help me implement the attached classroom heart model. First explain the variables, units, and assumptions in plain language. Then implement one baseline case with clearly labeled parameters and plots. Use the equations exactly as supplied, keep electrical time units in milliseconds and contraction/pump time units in seconds, and handle input pulse start and stop times explicitly. Help me check a known limiting case and numerical accuracy before adding comparison scenarios. Explain each code section so I can modify it myself. Label simplified outputs accurately; do not present passive voltage spread as a cardiac action potential or a force sum as cardiac output.”

Questions to discuss

- What does each model help you understand about cells working together?
- Which results agreed with your prediction? Which surprised you?
- What did the coding tool produce that you needed to check or correct?
- Which biological mechanisms are missing, and how might they change your conclusions?

Suggested reading

OpenStax Anatomy and Physiology 2e, Section 19.1, Heart Anatomy https://openstax.org/books/anatomy-and-physiology-2e/pages/19-1-heart-anatomy

OpenStax Anatomy and Physiology 2e, Section 19.2, Cardiac Muscle and Electrical Activity https://openstax.org/books/anatomy-and-physiology-2e/pages/19-2-cardiac-muscle-and-electrical-activity

Optional practice; no separate submission required.